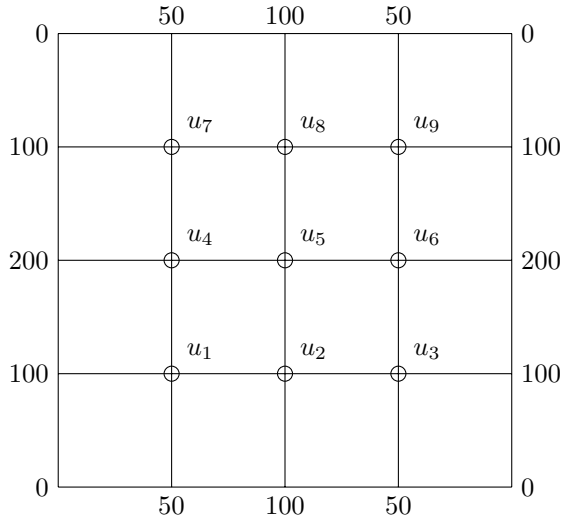
**Q.8.** Use Liebmann's iteration formula to solve Laplace equation $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ at the interior mesh points of the square region with boundary values given in the following figure.




[Take 2 iterations and obtain the result correct up to three decimal places.]